Data & AI

11. Inside ResNet18

Dasaem Jeong · Art & Technology, Sogang University

Last Time

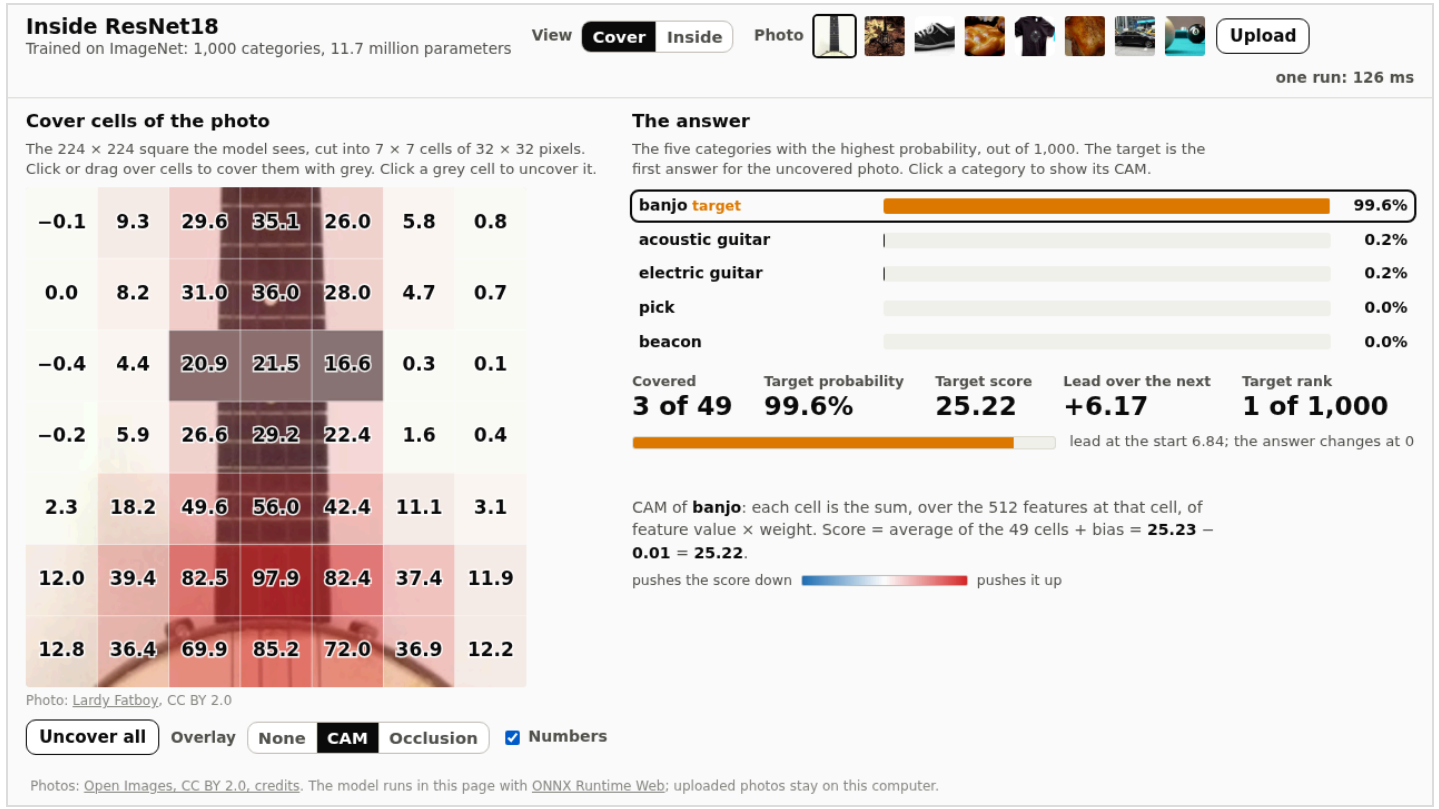
- A small CNN: convolution, max pooling, then 784×10
- ResNet18: 20 convolution layers, 11,689,512 parameters

Today

- Inside ResNet18: what changes its answer, and why
- Gradient ascent on the pixels: feature visualization, DeepDream, style transfer
- Adversarial examples

Inside ResNet18

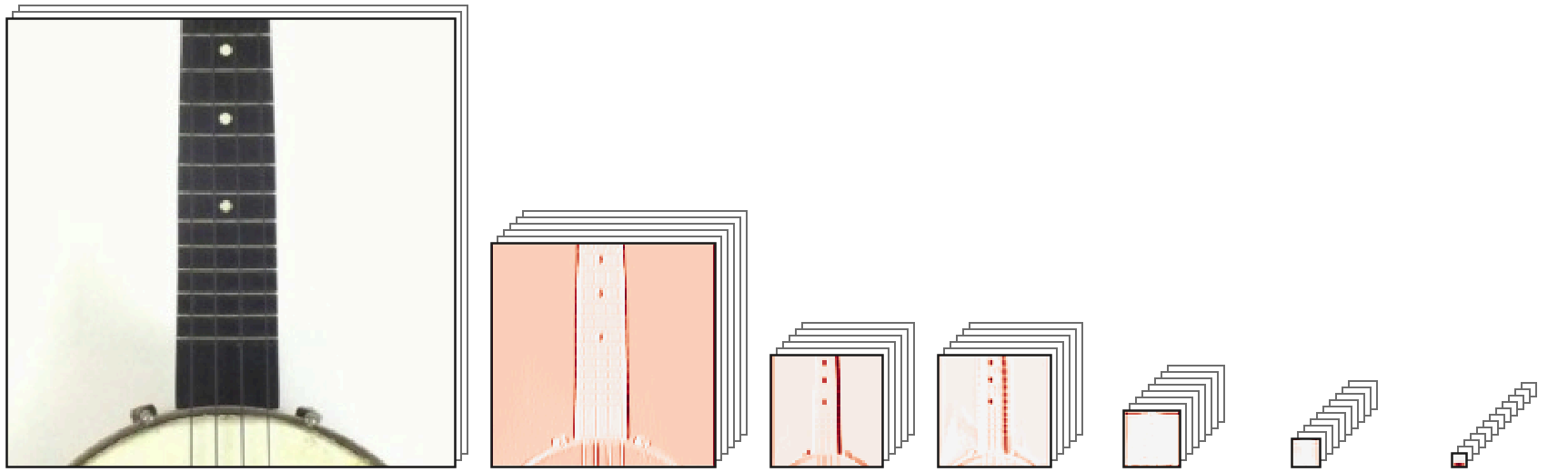
jdasam.github.io/mas1004/demos/lecture11-resnet18-cam



The Structure of ResNet18

ResNet18 Layer by Layer

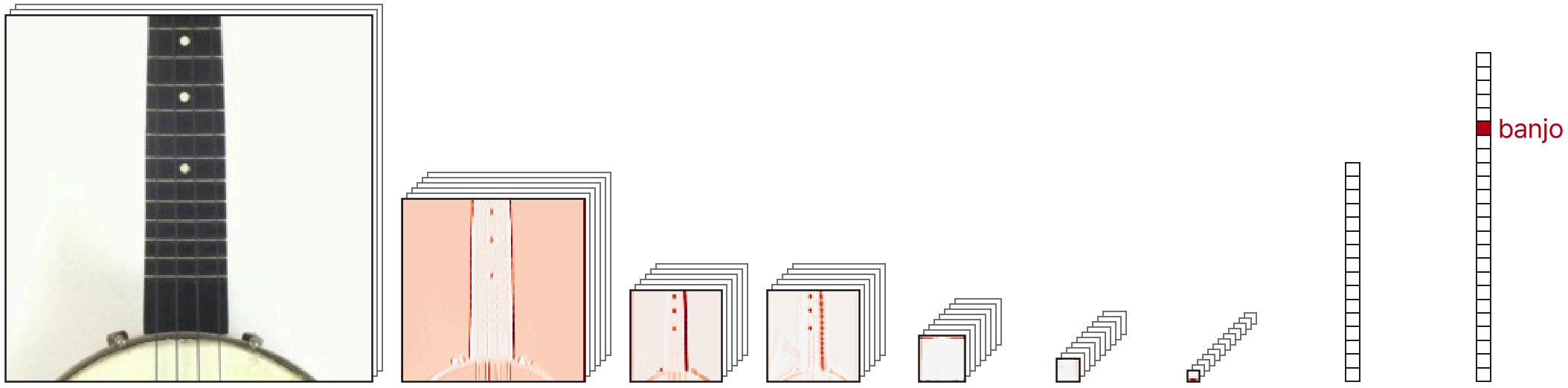
Output shape after each stage, for one 224 × 224 photo



input	conv1	max pool	layer1	layer2	layer3	layer4
224 × 224	112 × 112	56 × 56	56 × 56	28 × 28	14 × 14	7 × 7
3 colours	64 maps	64 maps	64 maps	128 maps	256 maps	512 maps

ResNet18 Layer by Layer

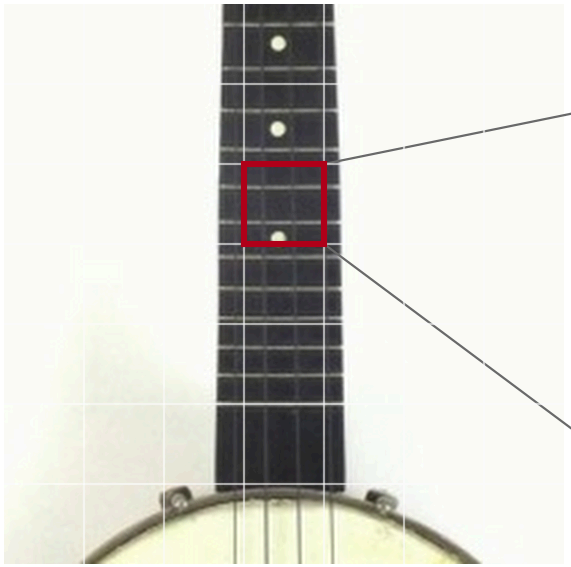
Output shape after each stage, for one 224 × 224 photo



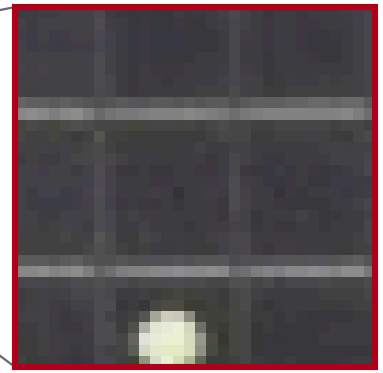
Receptive Field

The pixels that can change one cell of layer4

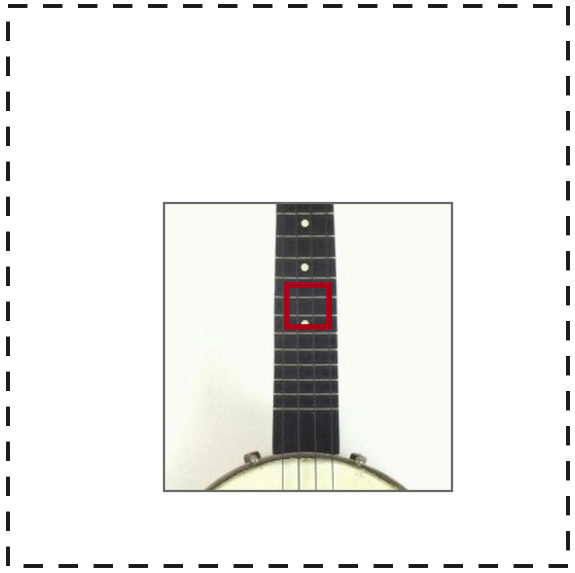
7 × 7 cells of 32 × 32 pixels



cell row 3, column 4

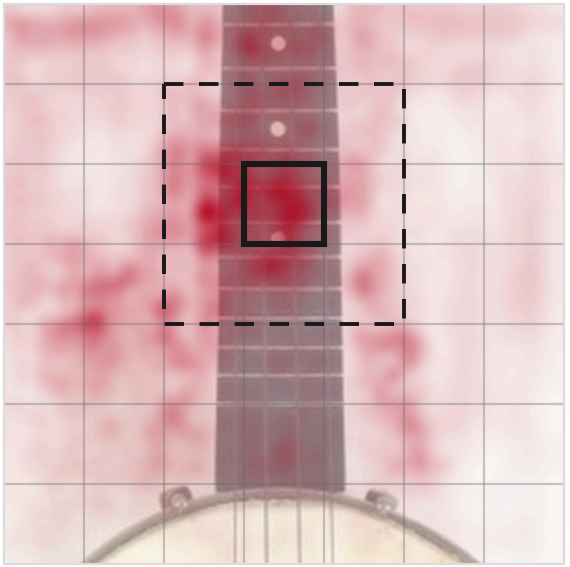


can reach it: 435 × 435 pixels



more than the whole photo (224 × 224)

how strongly each pixel moves it



8% inside the cell, 33% inside the 3 × 3 ce

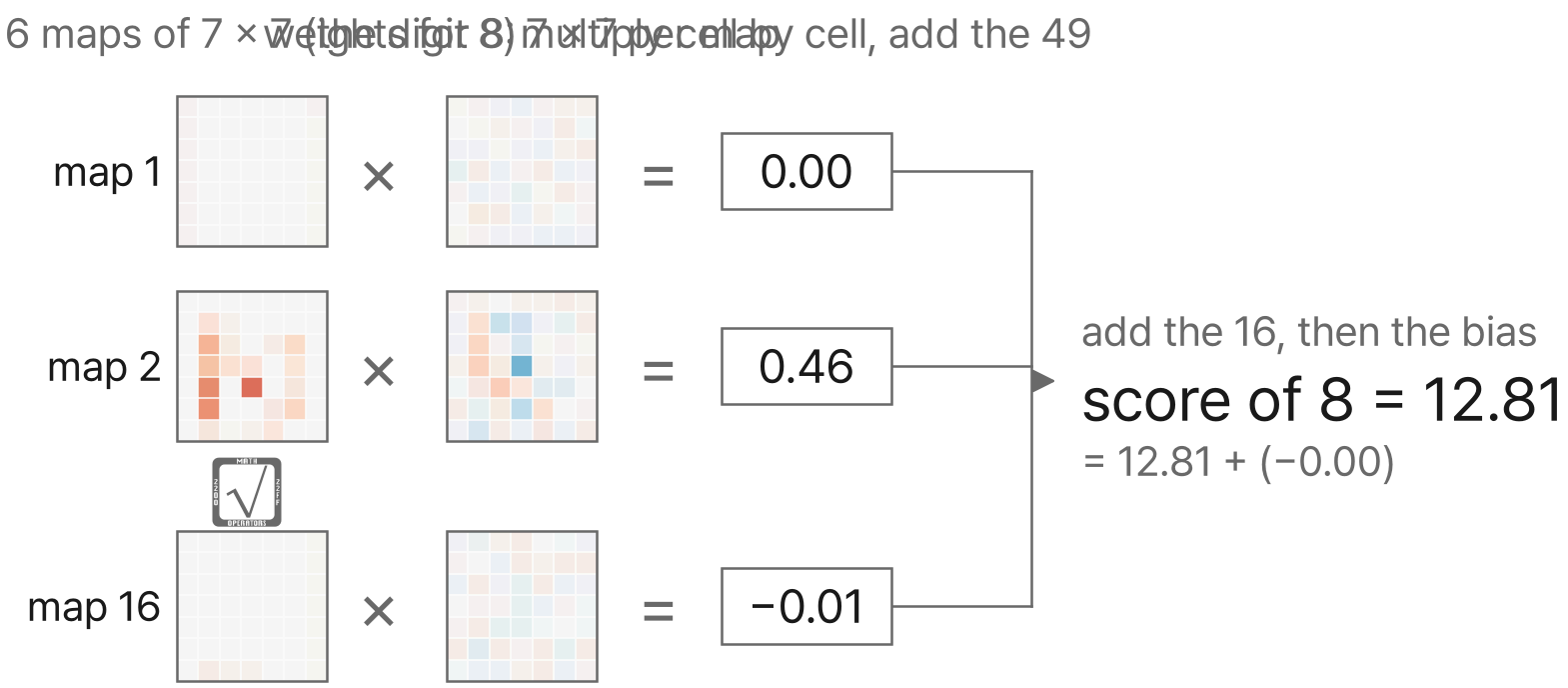
Residual Block

He, Zhang, Ren, Sun (2016), Deep Residual Learning for Image Recognition, CVPR

- Output = F(x) + x: two convolutions compute F(x), and the input x is added back
- ResNet18: 8 blocks of two 3 × 3 convolutions

The Last Layer of the Small CNN

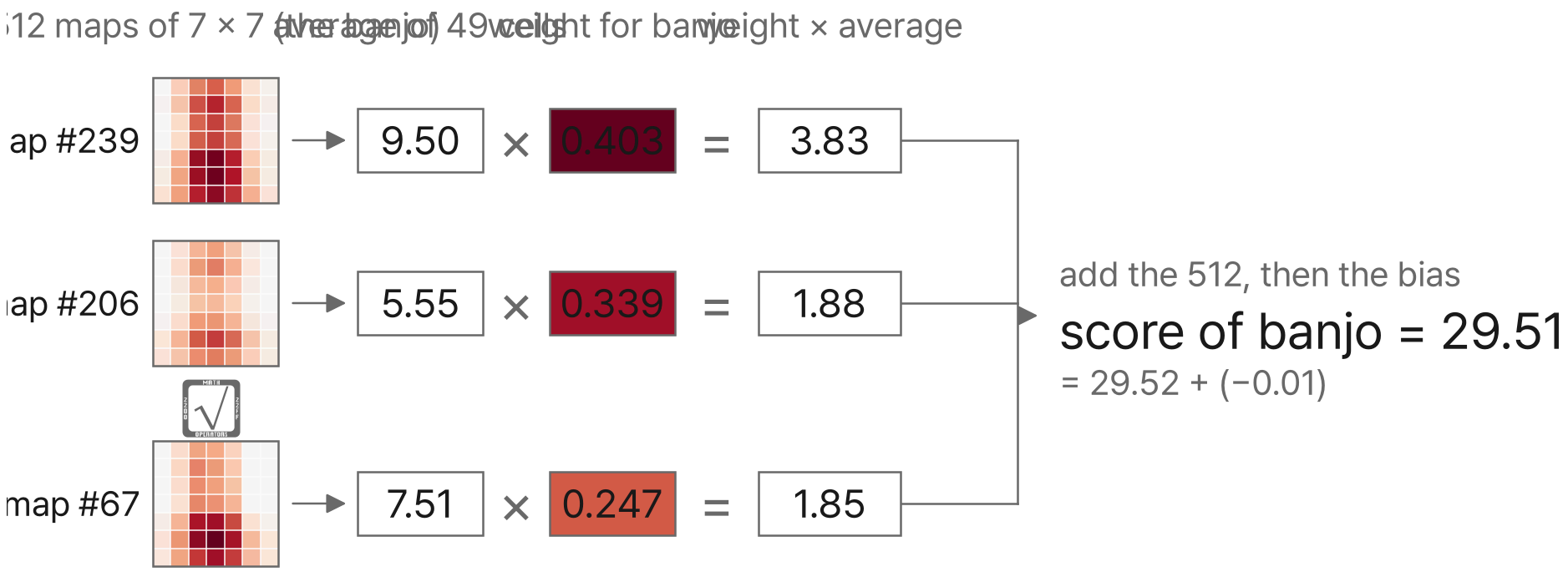
Lecture 9: 16 maps of 7 × 7, flattened into 784 numbers, then 784 × 10



Flattened: 16 × 7 × 7 = 784 numbers, each with its own weight. 784 weights × 10 digits = 7,840 weights

Global Average Pooling

ResNet18: each 7 × 7 map averaged into one number, then 512 × 1,000



Averaged: 512 maps become 512 numbers, one weight each, the same for all 49 cells. 512 weights × 1,000 categ

Splitting the Score by Cell

$f_k(c)$: value of feature k at cell c · w_k: its weight for the category

$$\text{score} = \sum_{k=1}^{512} w_k \cdot \frac{1}{49} \sum_{c=1}^{49} f_k(c) \; + \; b$$

Splitting the Score by Cell

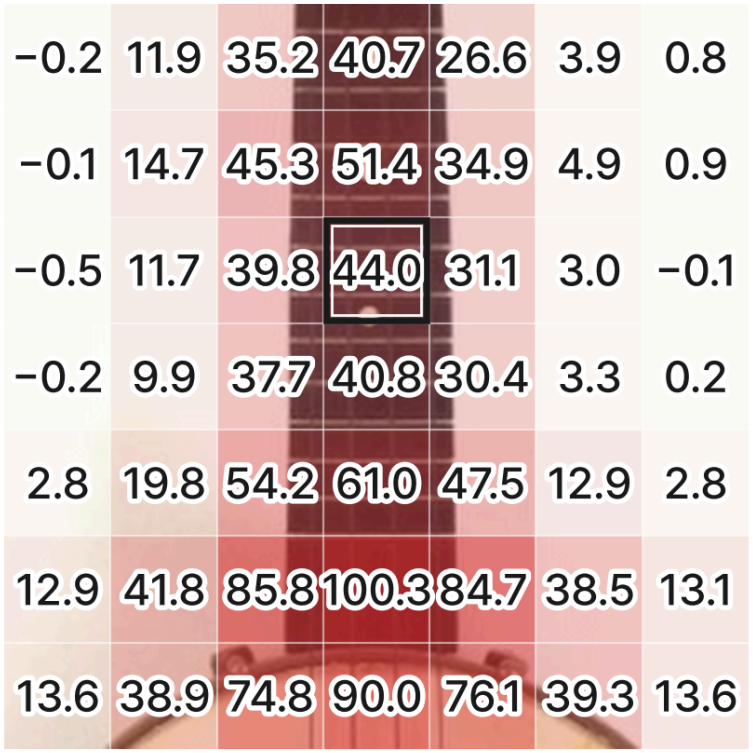
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$$= \frac{1}{49} \sum_{c=1}^{49} \underbrace{\sum_{k=1}^{512} w_k f_k(c)}_{\text{value of cell } c} \; + \; b$$

Class Activation Map

Zhou, Khosla, Lapedriza, Oliva, Torralba (2016), Learning Deep Features for Discriminative Localization, CVPR



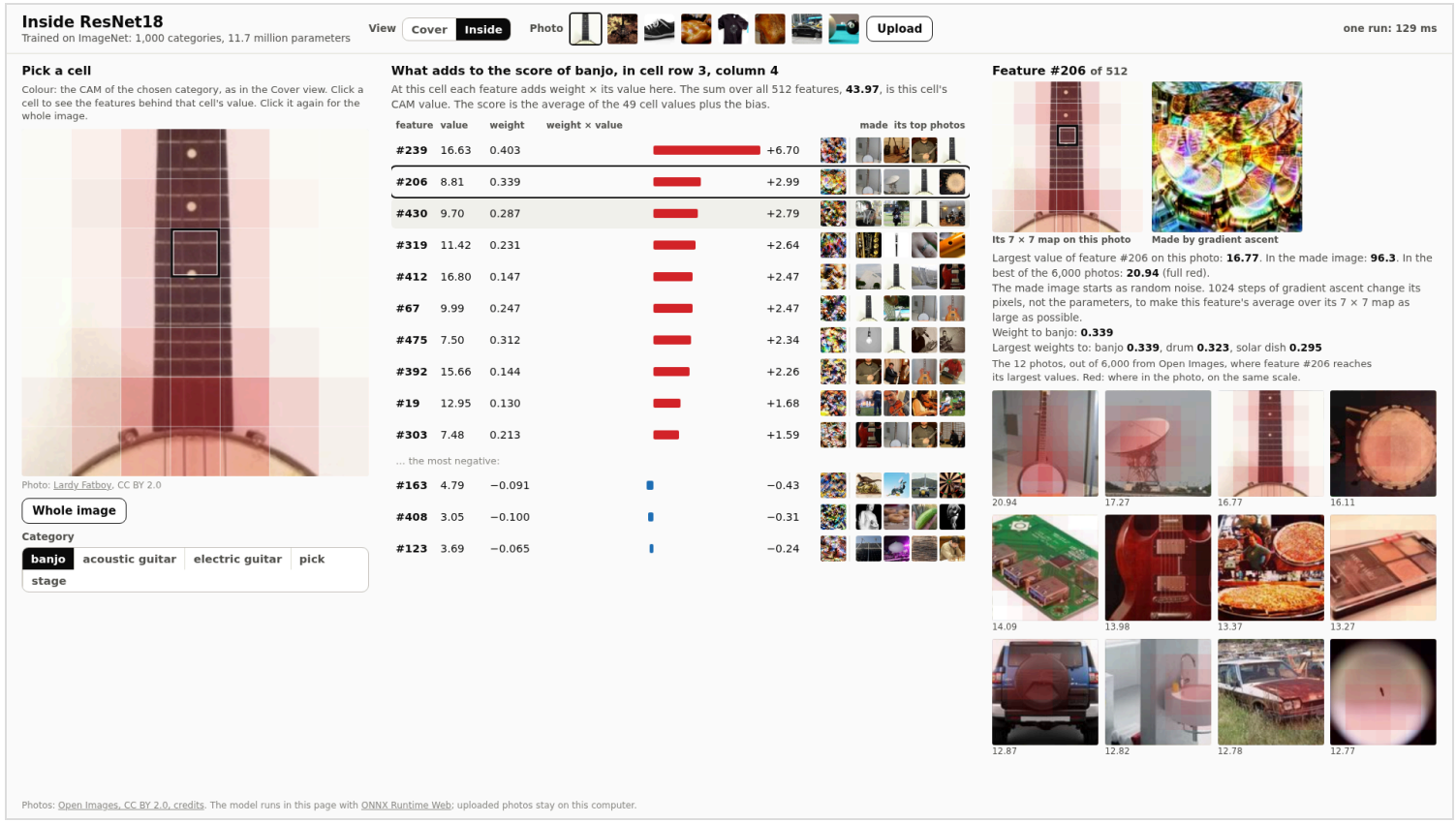
In each cell: weight × value, added over the 512 features

average of the 4[illegible]9.52	ls
bias	−0.01
score of banjo	29.51

pushes the score down ... pushes it up

Splitting a Cell by Feature

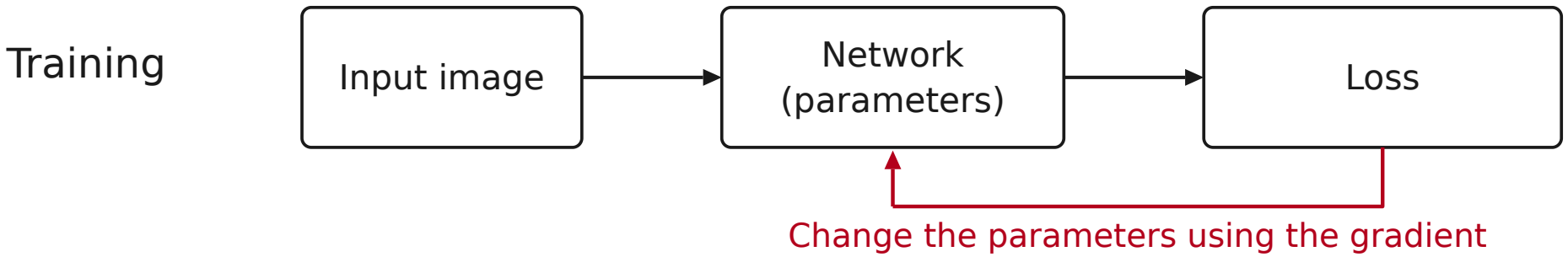
Inside view: banjo, cell at row 3, column 4



Gradient Ascent on the Pixels

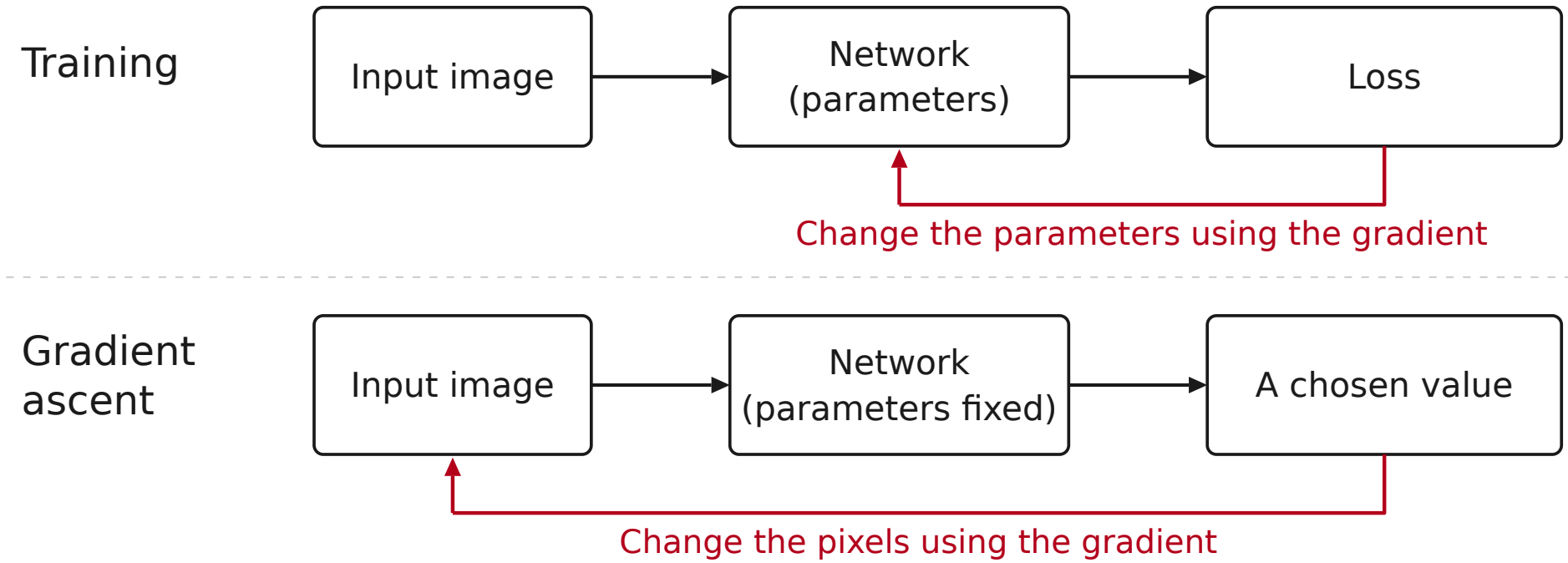
Gradients for the Pixels

Backprop gives a gradient for every input number too



Gradients for the Pixels

Backprop gives a gradient for every input number too



Feature Visualization

Olah, Mordvintsev, Schubert (2017), Feature Visualization, Distill



- Start from random noise, change the pixels so that one feature's average gets larger

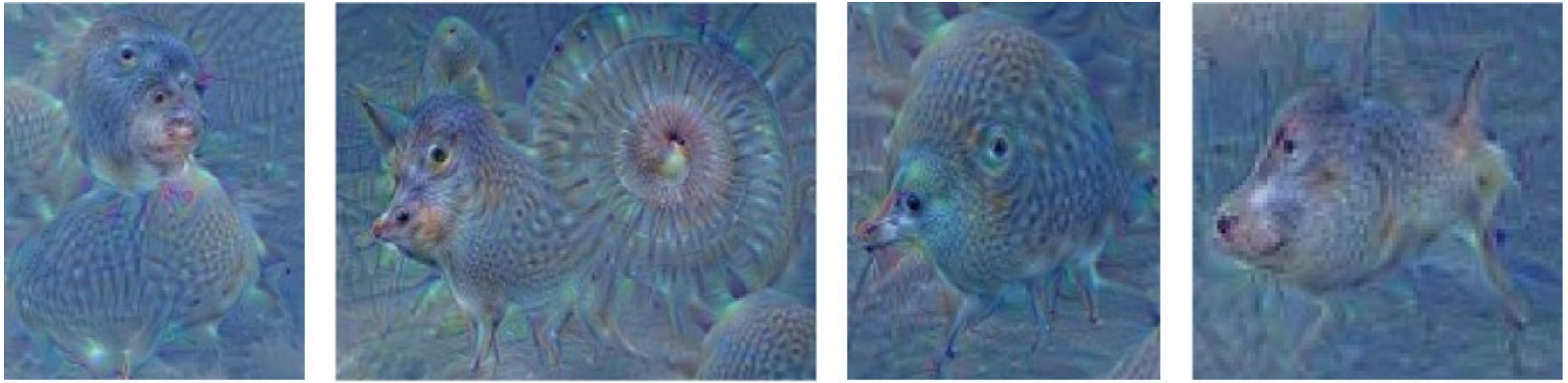
Class Visualization

Mordvintsev, Olah, Tyka (2015), Inceptionism: Going Deeper into Neural Networks, Google Research Blog



DeepDream

Mordvintsev, Olah, Tyka (2015), Inceptionism: Going Deeper into Neural Networks, Google Research Blog



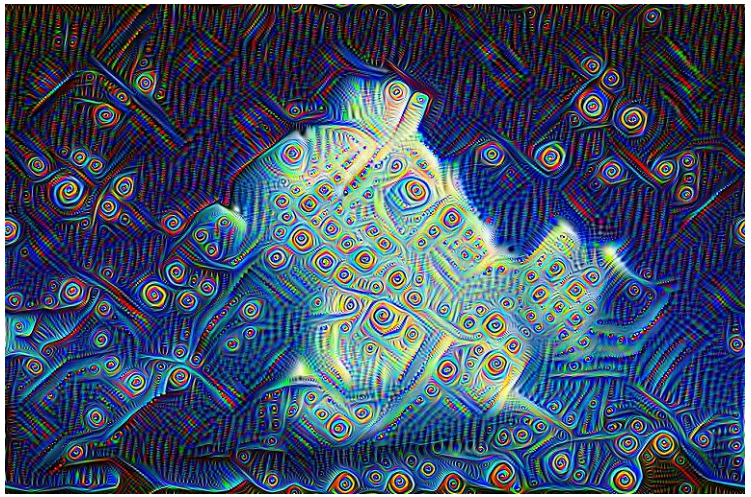
- Start from a photo, and make the values of a whole layer larger

DeepDream by Layer

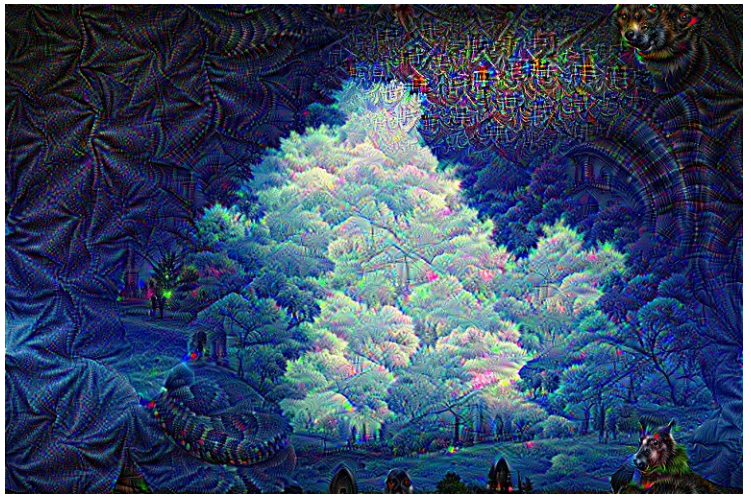
GoogLeNet, the same photo



Input



Shallow layer (inception3a)

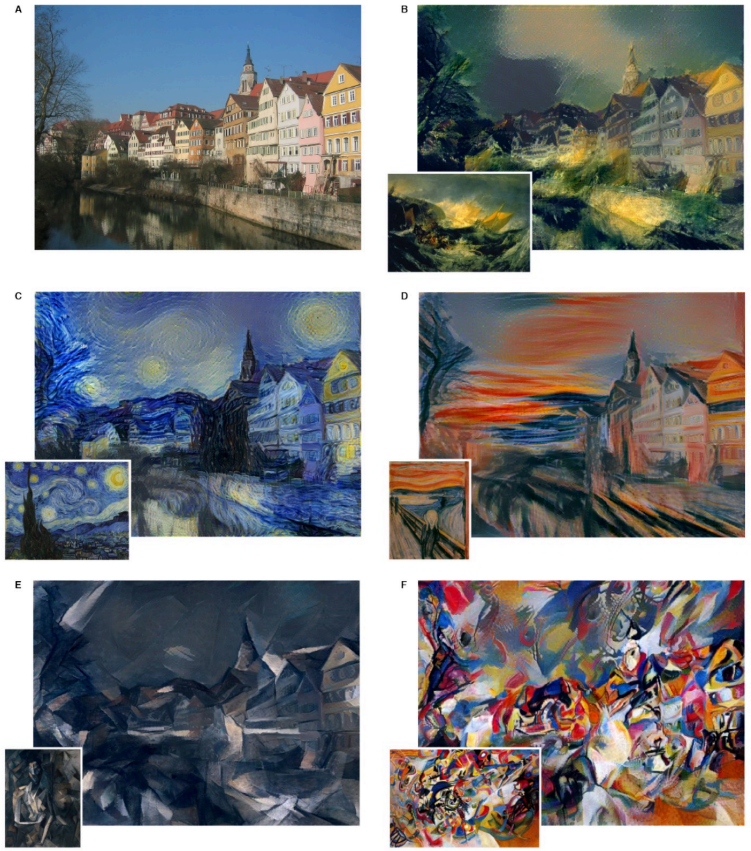


Deep layer (inception4c)

Input: Ralph F. Kresge, Cumulus6 (NOAA, public domain)

Neural Style Transfer

Gatys, Ecker, Bethge (2016), Image Style Transfer Using Convolutional Neural Networks, CVPR



- Content loss: the maps of a deep layer, cell by cell
- Style loss: which features appear together, position thrown away
- Change the pixels to lower the sum of the two losses

Style Transfer Result

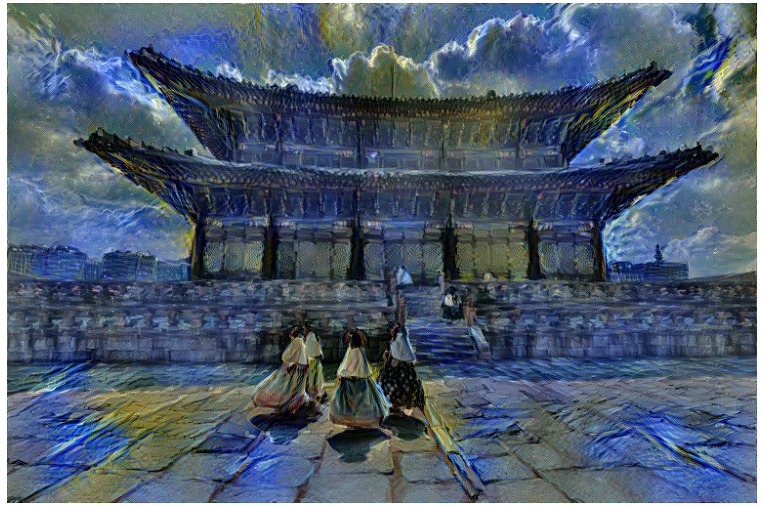
VGG19 · Gyeongbokgung photo and The Starry Night



Content



Style



Result after 400 steps from noise

Content: chsyang, Gyeongbokgung Geunjeongjeon (Public Domain Mark) · Style: Vincent van Gogh, The Starry Night

Adversarial Examples

Adversarial Examples

Goodfellow, Shlens, Szegedy (2015), Explaining and Harnessing Adversarial Examples, ICLR

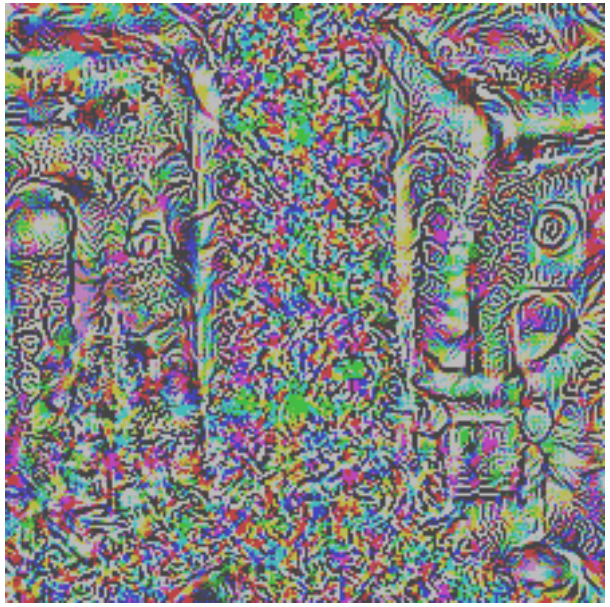
- The same gradient for the pixels, used to change the answer
- Every pixel moves by a tiny amount that a person cannot see

An Adversarial Banjo

ResNet18, the banjo from the demo, each pixel changed by at most 1 level out of 255



banjo 99.8%



Change × 60



toaster 100%

Many Small Changes Add Up

3 × 224 × 224 = 150,528 input numbers

- Each number changes by at most 1 out of 255
- Most of the 150,528 numbers move in the direction that raises the toaster score

Coming Up

- Assignment 1 due Friday, October 16, 22:00